

End-term Report: River Network Segmentation Project

Srinivasan M (IMT2021058) Siddharth Kothari (IMT2021019)
Sankalp Kothari (IMT2021028)

December 14, 2025

1 Summary of Accomplishments

In this semester, we primarily worked on running three new models with the uncorrupted data - DeepLabV3 [1], Segformer [2] and Maskformer [3]. The results from these were going to be used as benchmarks for our dataset, to evaluate the quality of the dataset, and to provide a SOTA comparison. We also worked towards making visualizations which were used to provide substance and necessary background to the paper. Each of these is described below.

1.1 Development of Codes and Training for DeeplabV3, Segformer and Maskformer

The first task which we worked on was the development of training and evaluation codes for the following models - DeepLabV3, Segformer and Maskformer. This was indeed a difficult task, especially for DeepLabV3. We tried out the following elements -

1. The first step was to try and use Prof. Ujjwal's own code (which used DeepLabV3+) which was used for his paper [4]. This did not work for our dataset, hence we had to abandon this approach.
2. The next step was to try and use the implementations for all three which were provided by the mmsegmentation repository [5] (<https://github.com/open-mmlab/mmdetection>), however this approach also did not work out as the config files required to work with the models was very complicated.
3. Finally, we had to use DeepLabV3 over DeepLabV3+ as that was the only model which worked completely, and had a readymade implementation in the torchvision library. This model was run with the uncorrupted data, and the corresponding results were obtained.
4. Segformer implementation was much easier due to the extensive documentation provided. The documentation helped to quickly write the required code (which remained unchanged for our implementation as well), and the model was run and results recorded.
5. Maskformer implementation was also relatively harder, due to no proper documentation and the fact that Maskformer worked on different image sizes. Changes had to be introduced to the codes to make them work with our data (512x512). This was, however, easier than DeeplabV3 and we were able to get it run soon, and recorded the qualitative and quantitative results.

A more detailed discussion of the results is provided in a later section.

1.2 Data Visualizations

One of the primary visualizations that we made was the visualization of the U-Net [6] architecture used by us. It is shown in figure 1. This is used to provide a visualization of the architecture as well as the corruption methodology, and was created via Lucidchart.

We also created a few codes which could read the csv files with the coordinates for each image, and make a shapefile out of them, with each image rectangle representing the coordinates of an image, which was then superimposed on a world map via QGIS. This was done to visualize the area which the dataset covers. There were 6 different areas, and we expanded on each of them, while also providing a broader map as shown in figures 2 and 3. These provided a detailed picture of the dataset cover. The codes were also used to determine the area of the amazon basin covered by the dataset (in sq. km) which has been included in the paper as well.

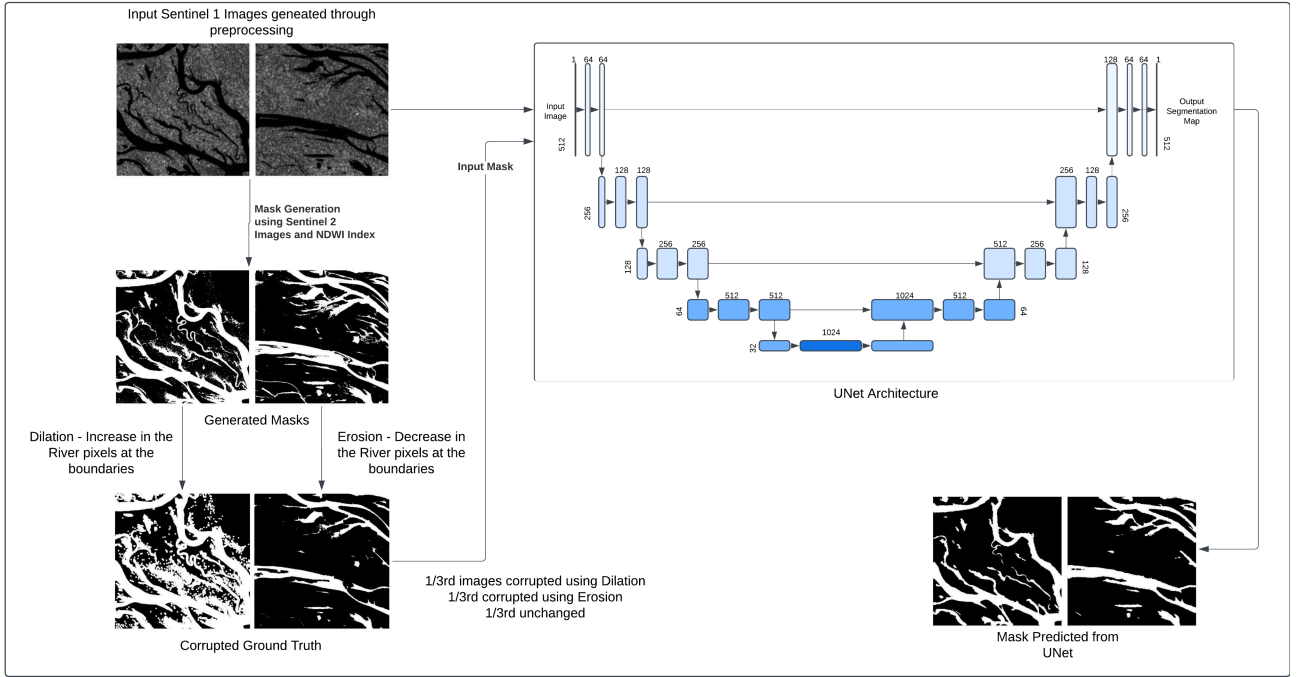


Figure 1: The workflow of our proposed method used for simulating *adversarial attacks* in ground truth generation for inland water body segmentation.

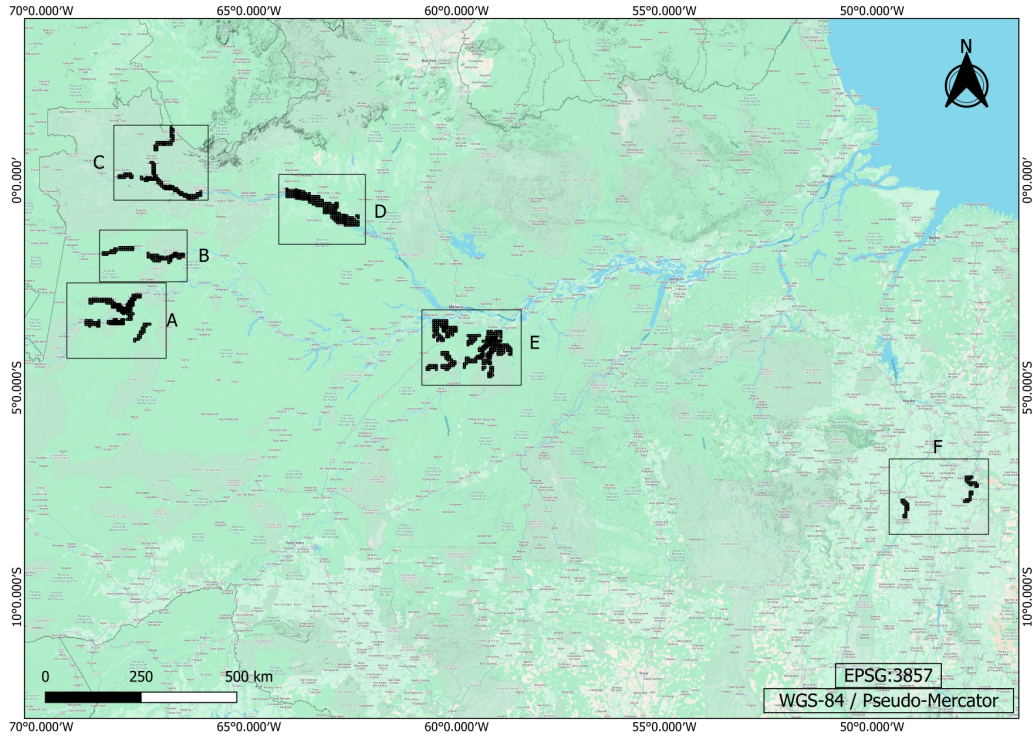


Figure 2: Regions of the Amazon River Basin used as images in our curated dataset for both training and testing. The images are taken from six non-overlapping regions, marked as regions A to F, capturing the complexity of the geometry of the river system. This image has been generated using QGIS.

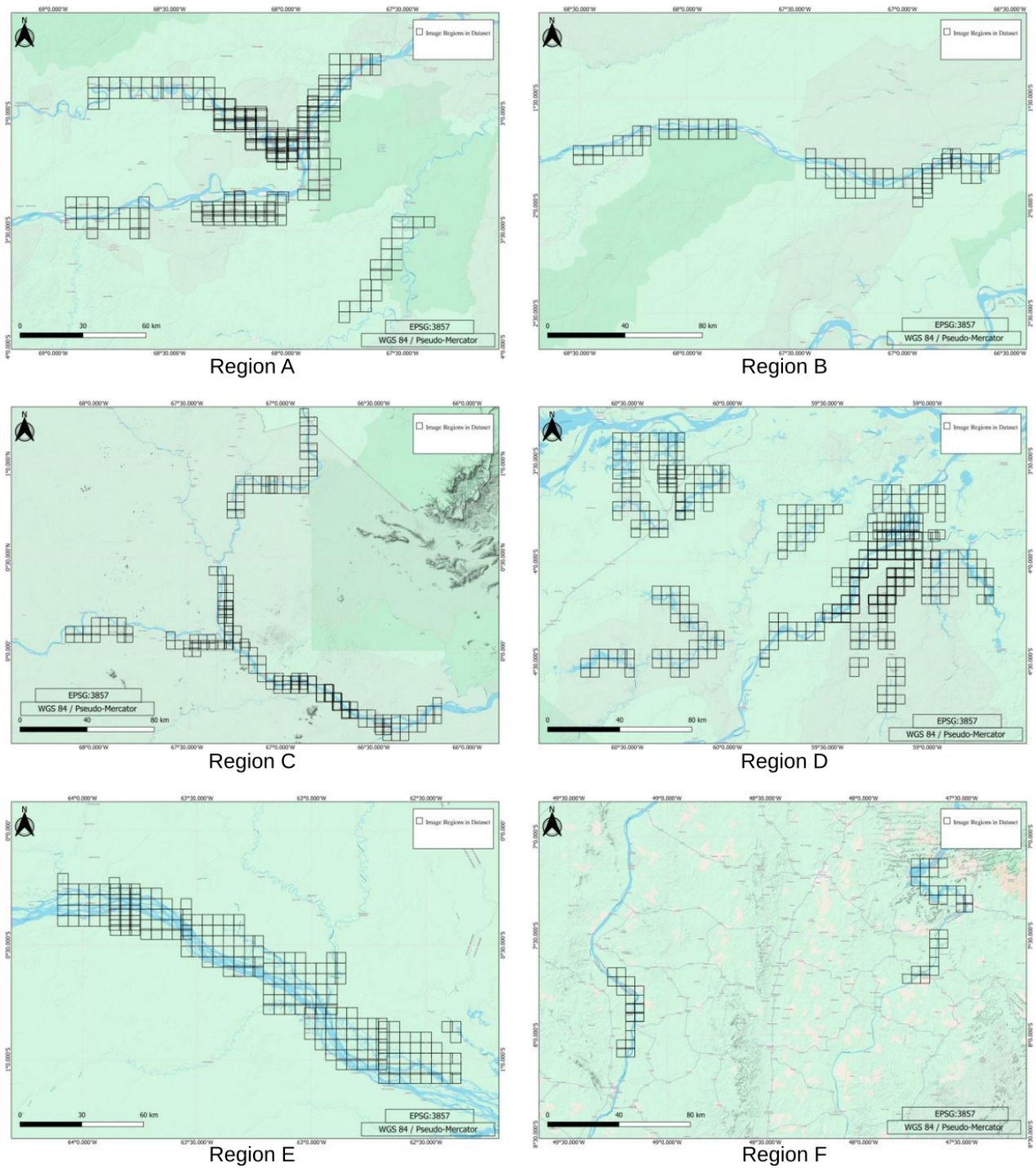


Figure 3: Zoomed-in version of regions A to F from Figure 2, where the rectangles indicate the images used in our curated dataset. These images have been generated using QGIS.

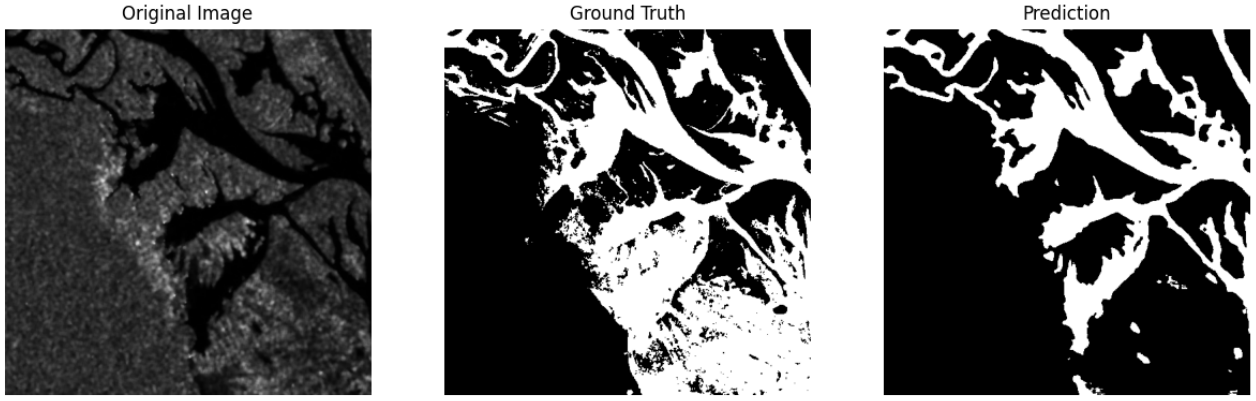


Figure 4: A data point where the ground truth is not fully accurate due to a non-optimal threshold used in ground truth generation.

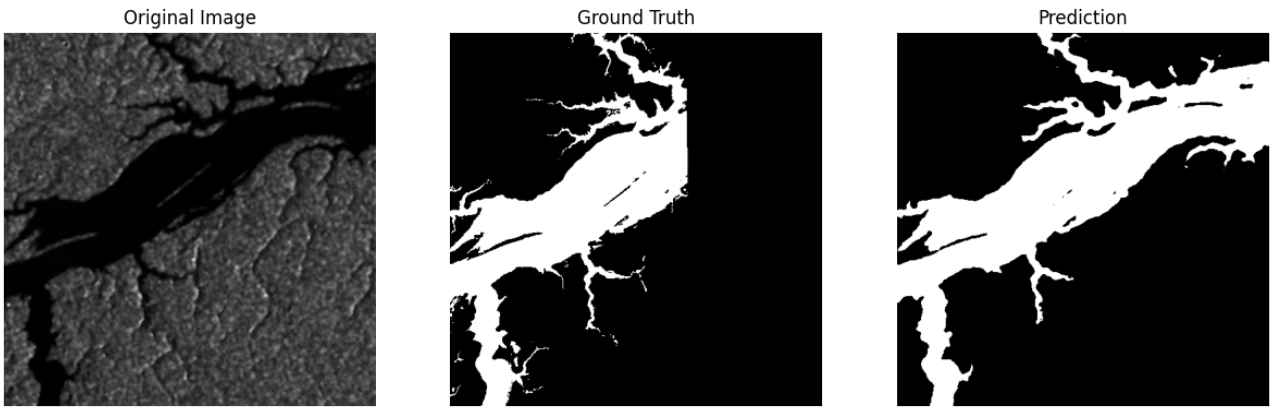


Figure 5: A data point where the ground truth is incorrect due to incomplete ground truth owing to the unavailability of the Sentinel-2 image in the specific region.

1.3 Identification of Incorrect Ground Truths

One of the steps that we carried out this semester was to analyze the images which have incorrect ground truths. The accuracy provided by the segmentation models all capping at 0.95 indicates that there are some images, which no matter what, cannot be classified correctly. We thus began to investigate which are these images and what are the types of errors observed in these images. There were broadly 2 types of errors observed; the first type was caused due to incorrect thresholding, an example of which is shown in figure 4; the second type was caused due to incomplete regions in the Sentinel-2 image, an example of which is shown in figure 5. These errors do indicate that the automated method too does require manual adjustments to be fully correct.

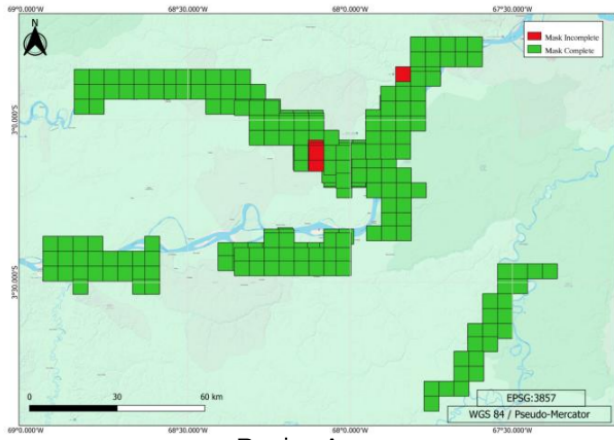
Hence, we also investigated manually each of the 1200+ images, and made a list of which images had seemingly incorrect labels. We also plotted it on the map to see whether the erroneous regions are grouped together, indicating the discrepancies being concentrated to only one region. This was the case for the most part, as can be observed in figure 6. This helped us to further accurately pinpoint the 2 reasons for these flaws.

2 Observations and Results

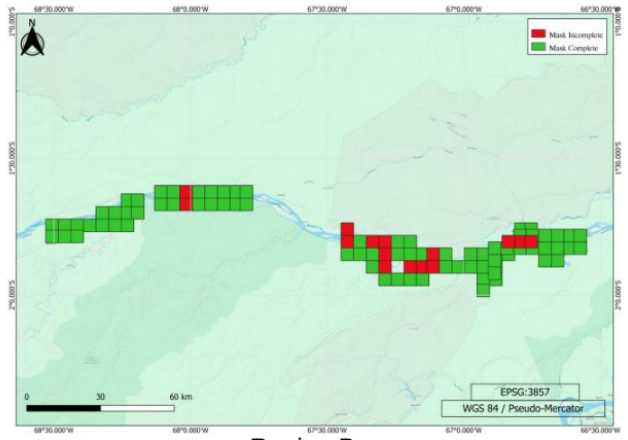
The results are summarised in Table 1. The results show that each of the models performs similar to U-Net on the uncorrupted data, and that sets a benchmark for the quality of this dataset.

3 Paper Rewriting

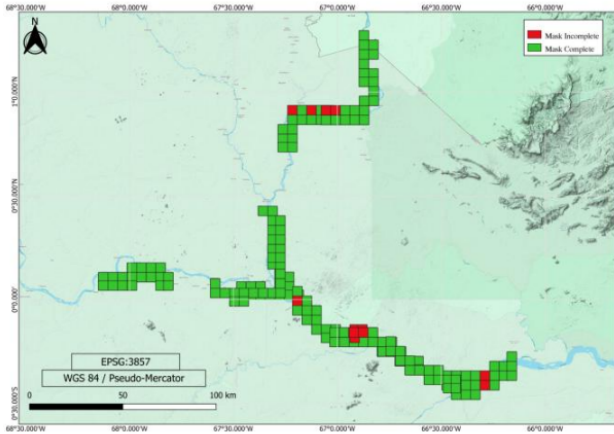
The following is a summary of the changes we made to the versions of the paper after observing new results and rewriting the paper based on them



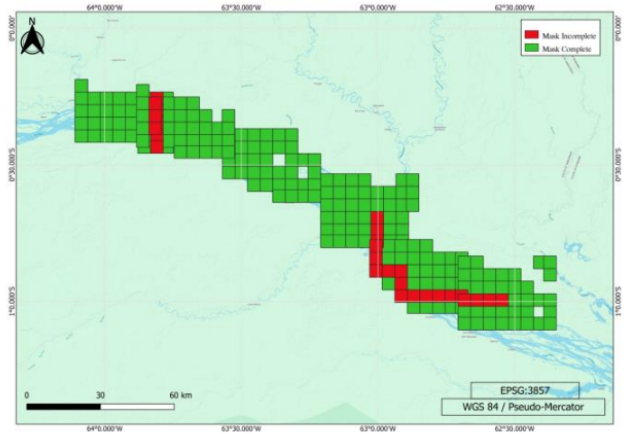
Region A



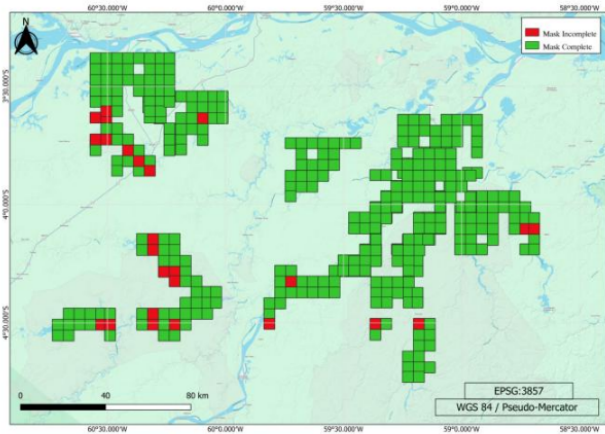
Region B



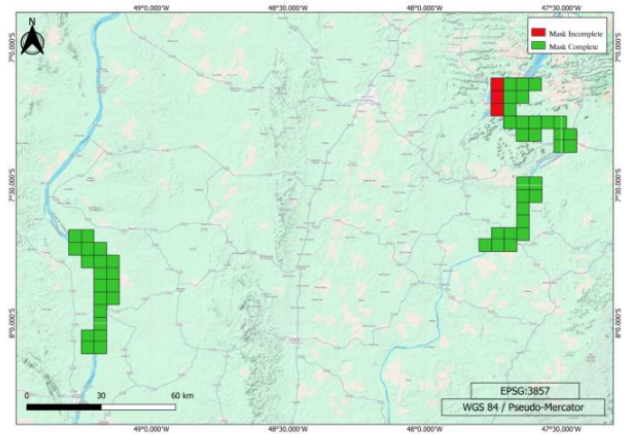
Region C



Region D



Region E



Region F

Figure 6: Images with accurate masks (green) and anomalous masks (red), where the latter is due to incomplete data in Sentinel-2 images for ground truth generation. Observing these masks superimposed on the Amazon River Basin helped identify the cause for this anomaly.

Table 1: Performance of the various models on uncorrupted/clean data.

Model	Dice Coeff.	IoU	Precision	Recall	F1 Score	Specificity	Accuracy
U-Net [6] (Baseline)	0.770	0.823	0.896	0.869	0.865	0.971	0.955
DeepLabV3 [1]	0.770	0.819	0.891	0.865	0.864	0.965	0.953
SegFormer [2]	0.770	0.829	0.908	0.868	0.870	0.975	0.957
Mask2Former [3]	0.774	0.827	0.894	0.872	0.869	0.973	0.957

1. **Introduction and Related Work:** We restructured the introduction to emphasize inland water body segmentation from SAR imagery, grounding the motivation in the Amazon River Basin and highlighting challenges from complex river geometries and SAR-specific noise. We sharpened the focus on CNN-based encoder-decoder models, particularly U-Net, and framed their vulnerability to noise as the core problem addressed. We introduced morphological label noise as a realistic simulation of human annotation errors within a data poisoning framework. To improve clarity, we separated background material into a dedicated Related Work section covering traditional, deep learning, dataset, and adversarial approaches, and refined the contributions to highlight our adversarial corruption pipeline and dataset release.
2. **Method and Experiments Section:** We restructured the method section to clearly separate dataset generation, model design, adversarial simulation, and implementation details. We formalized a two-stage dataset pipeline combining Sentinel-1 SAR preprocessing with Sentinel-2 NDWI-based mask generation and quantified regional coverage across the Amazon Basin. We added spatial visualizations and standardized preprocessing steps to improve clarity and reproducibility. We expanded experiments to include multiple state-of-the-art segmentation models while retaining U-Net as the robustness benchmark. We refined the corruption strategy using probabilistic, data-dependent morphological noise and added implementation and release details for transparency.
3. **Results and Discussion Section:** We expanded the results from a U-Net-only analysis to a comparative evaluation with DeepLabV3, SegFormer, and Mask2Former on clean data to assess dataset quality and performance saturation. We separated clean-data benchmarking from adversarial experimentation to improve narrative flow. We characterized erosion-induced corruption saturation using box plots and SSIM analysis and added extensive qualitative evaluations to study robustness and failure modes. We analyzed model behavior before and after thresholding and identified inadvertent label noise due to NDWI thresholding and incomplete Sentinel-2 coverage. Together, these additions unify quantitative trends, qualitative evidence, and dataset-level diagnostics.

4 Conclusion

1. We implemented and benchmarked three state-of-the-art segmentation models—DeepLabV3, SegFormer, and Mask2Former—on uncorrupted data to validate the quality of our curated Amazon River Basin dataset and establish strong performance baselines against U-Net.
2. We developed multiple visualizations to support the paper, including architectural workflow diagrams, dataset coverage maps using QGIS, and region-wise zoomed analyses, enabling clearer communication of the methodology and spatial extent of the dataset.
3. We systematically analyzed ground truth errors by manually inspecting over 1200 images, identifying inaccuracies arising from NDWI thresholding and incomplete Sentinel-2 coverage, and mapped these anomalies spatially to understand their regional concentration.
4. We rewrote and restructured the paper to reflect new experimental insights, improving clarity in the introduction, method, and results sections, expanding comparative evaluations, and strengthening the narrative around adversarial label noise, dataset construction, and robustness analysis.

References

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